Evaluating Data Centers as Local Energy System Assets: A System-Integrated PUE Approach

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Abstract

Data centers are growing electricity loads, but they can also export waste heat, provide flexible demand, and use storage to reduce local network stress. Power Usage Effectiveness (PUE) treats the electricity used for those services as overhead, so an integrated site can appear less efficient even when it creates external system value.

This paper defines System-Integrated PUE (siPUE), a companion indicator that adds a single delay-adjusted, energy-equivalent credit to the PUE numerator, drawn from the combined heat, flexibility, peak, and carbon-damage value of a jointly optimized design. The metric is tested with a two-stage design and operation model for four data-center archetypes and ten shared uncertainty states, using representative-day operation, a stylized three-node DC-OPF network, heat-recovery constraints, lifecycle delay accounting, and land-cost accounting.

The reference case is the services-disabled comparator, which keeps the same 15% efficiency retrofit and physical constraints as the service-active cases. Weighted by annual IT energy, PUE and siPUE are both 1.167 for this comparator. Relative to it, the baseline service-active design raises PUE to 1.238 but lowers siPUE to 1.139, with 5.603 EUR/MWh IT of delay-adjusted system benefit. The robust design raises PUE to 1.343 and lowers siPUE to 1.152, with 10.825 EUR/MWh IT of benefit. The service-active signal is therefore material but not overstated: credits are about 1.40 and 1.08 times the baseline and robust facility-energy burdens, respectively. Excluding carbon damage reverses the average siPUE improvement, so the result is conditional on the stated social-carbon valuation. Internal checks confirm

metric consistency; external validation requires observed project data.

Keywords: Data centers, Power Usage Effectiveness, System-Integrated PUE, waste-heat reuse, flexibility, robust optimization, local energy systems

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1. Introduction

Electricity demand from European data centers has become a material planning variable for transmission and distribution networks. Commission estimates put EU consumption at 76.8 TWh in 2018, with projections of 98.5 TWh by 2030 under a trend case and up to 160 TWh under a high-growth case [1]; more recent assessments suggest roughly 96 TWh in 2024 and nearly 168 TWh by 2030 [2]. Growth is reinforced by traffic expansion and compute-intensive AI workloads, which change not only annual demand but also operational ramps, peaks, heat rejection, cooling needs, and flexibility headroom [3, 4].

At this scale, data-center build-out becomes a local infrastructure and investment problem. Connection rights, congestion, heat offtake, flexibility-market eligibility, land-use approvals, permitting timelines, and community acceptance increasingly determine which projects proceed [5, 6]. The EU energy transition adds a second pressure: electricity and heat systems need flexibility, digital coordination, and sector coupling [7, 8], while sustainability reporting for large data centers is being formalized through Directive (EU) 2023/1791 and sector initiatives [9, 10]. Data centers can therefore act either as inflexible loads or as service-active assets that export waste heat, provide grid support, and manage dynamic AI-driven loads [11, 12].

The reporting problem is that PUE evaluates only internal electricity overhead. Heat pumps, heat exchangers, controllability, storage, and telemetry can increase facility electricity even when they reduce local system cost. This creates the KPI gap studied here: PUE observes the facility-energy burden, while the independent system-cost benefit remains outside the metric. siPUE is introduced as a companion indicator, expressed in PUE's own energy units, that translates that benefit into the same numerator language as PUE while preserving PUE's internal-efficiency role.

1.1. Related Work

The relevant literature spans data-center metrics, heat recovery, flexibility, integrated energy systems, and robust decision-making. PUE remains

the dominant facility-efficiency ratio [13, 14], but critiques show that it omits flexibility, source effects, climate dependence, and wider sustainability dimensions [15, 16]. Parallel indicators such as CUE, WUE, ERF/ERE, and REF add specific dimensions through the ISO 30134 family and EU reporting practice, but they do not provide a market-valued PUE numerator credit.

Waste-heat and district-heating studies establish data-center heat as a real system service [11, 17], while demand-response and flexibility studies show that IT load, cooling, and batteries can provide grid services under suitable market rules [18, 19]. Integrated energy-system papers quantify these services inside optimization models and show potential cost or emissions reductions [20, 21]. These studies motivate system valuation, but they leave open the standalone reporting question: how should an operator report a service-active design when PUE rises because the service itself uses electricity?

Broader scoring approaches, including DESS and DCRE, widen the resource and context boundary [22, 23], and recent heat and flexibility reviews reinforce the need for quantifiable energy-system indicators [24, 18]. The gap is not that local-system value is unknown; it is that current reporting metrics do not feed the combined, delay-adjusted heat, flexibility, peak, and carbon-damage value of a jointly optimized design back into a PUE-normalized signal.

The two-stage model below uses an established robust-optimization structure as the uncertainty testbed. Minimizing worst-case regret follows the minimax-regret tradition [25, 26]; the secondary near-worst-case mean-cost tie-break is a lexicographic scalarization [27, 28]. The methodological claim is therefore limited: the model is used to observe how PUE and siPUE respond to the same robust design choice, not to propose a new robust-optimization criterion.

1.2. Synthesis and Gap Identification

Table 1 positions the proposed metric against the closest reporting metrics and integrated-system modeling literature. The contribution lies in the accounting translation from local-system value into a PUE numerator credit.

Across this literature, three persistent gaps remain:

- Fragmentation: The reporting metrics reviewed here (PUE, CUE, WUE, ERF/ERE, REF, DESS, DCRE) each measure one dimension or resource bundle, leaving local-system services without a companion score expressed in PUE's own energy units

Table 1: Comparison of siPUE with adjacent data-center metrics and modeling approaches.

Approach	Primary function	Values local-system services?	PUE numerator credit?	Gap relative to siPUE
PUE [13, 29]	Facility electricity divided by IT electricity	No	Baseline ratio	Treats heat-pump, storage, and control electricity as overhead even when they create external value.
ERF/ERE [29]	Reused-energy reporting and energy-reuse effectiveness	Heat only, in energy terms	Partial reuse adjustment	Omits flexibility, peak, network, carbon-damage, and delay effects.
CUE [15]	Carbon emissions per IT energy	Carbon footprint only	No	Parallel carbon-intensity indicator that stays outside the PUE numerator-credit boundary.
WUE [29]	Water consumption per IT energy	No	No	Tracks water use outside the energy-system value boundary.
REF [29]	Renewable-energy share	No	No	Tracks procurement mix; local heat, flexibility, peak, and carbon-damage value, adjusted for construction delay, remain outside its boundary.
DESS [22]	Broader sustainability score	Partial, contextual	No	Qualitative and contextual metric that stays outside the PUE numerator-credit boundary.
DCRE [23]	Resource-effectiveness view of energy, water, water stress, and climate context	Indirect	No	Broader resource metric outside PUE numerator-credit reporting.
Integrated energy-system optimization papers [20, 21]	System-cost or emissions optimization	Yes, inside the model	No	Quantifies system impacts while leaving standalone reporting-compatible numerator credits undefined.
siPUE (this study)	Companion indicator for service-active data centers	Yes: heat, flexibility, peak, and carbon damage, adjusted for construction delay	Yes	Converts the independent system-cost benefit into an energy-equivalent numerator credit and reduces to PUE when no services are active.

- Lack of market valuation: Heat reuse, flexibility, and carbon impacts are rarely valued using empirical electricity, heating, reserve or ETS price data; most studies rely on engineering assumptions or ad hoc pricing

- Absence of a numerator credit for jointly optimized local-system value: Among the approaches reviewed here, no metric was identified that feeds the combined, delay-adjusted value of a jointly optimized design's heat, flexibility, peak, and carbon-damage effects back into the PUE numerator, expressed in PUE's own energy units

1.3. Conceptual Framework and Research Questions

The paper is structured around a conceptual framework that links energy characteristics, data center design decisions, and local energy system outcomes. Evidence from policy documents, market data, and industry practice provides qualitative and institutional context for the energy characteristics and business readiness layer, while the model formalizes the data center responses and quantifies system outcomes. Section 2 is therefore treated as contextual synthesis rather than a standalone structured-review result. This leads to the following model-facing research questions:

1. RQ1: Can a companion indicator, expressed in PUE's own energy units, reveal the combined local-system value that a jointly optimized design creates through heat, flexibility, peak, and carbon damage, adjusted for construction delay, while reducing exactly to PUE when no services are active?

2. RQ2: How do robust, cost-minimizing design decisions shift traditional efficiency metrics and system-integrated indicators?

The remainder of the paper first provides the contextual synthesis, and then addresses RQ1–RQ2 through the metric definition and model-based analysis.

1.4. Contributions

This paper makes four contributions:

1. It integrates evidence on data centers and local energy systems, emphasizing how energy characteristics, AI workload dynamics, and integration options shape local business impacts and investment readiness.

2. It defines and tests siPUE as an accounting bridge between PUE's facility-energy burden and the combined local-system benefit of a jointly optimized design.

3. It uses a two-stage optimization model with scenario-perfect benchmarks and a shared normalized min–max regret design as the uncertainty testbed for evaluating the metric, with lifecycle delay, carbon ledgers, land accounting, heat recovery, flexibility, storage, and DC-OPF network constraints.

4. It evaluates siPUE for four archetypes and ten shared uncertainty states, reports the PUE–siPUE value-channel result, decomposes the siPUE credit by ledger component, and tests sensitivity to the reference price, heat price, flexibility price, cooling overhead, land availability, and construction delay.

2. Data Centers in Local Energy Systems — Policy and Market Context

This section summarizes the policy and business context for integrated data centers, building on the demand and workload trends already established in Section 1. Decarbonization, electrification, and sector coupling increase the value of flexible demand, heat reuse, and digital coordination [7, 8]. The business question is whether those services are investable: integration can add revenue and reduce grid-access risk, but it also adds capital cost, contract risk, and exposure to local infrastructure timing.

The section uses a macro–meso–micro lens informed by the multi-level perspective on socio-technical transitions [30]. Macro policy sets reporting, carbon, funding, and market-design incentives; meso infrastructure determines grid, heat-network, land, and permitting feasibility; micro finance converts those conditions into contracts, cost of capital, and project timing. This framing motivates why a reporting-compatible metric must account for externally valued services without abandoning PUE comparability. Figure 1 summarizes this structure and the cross-level conditions that jointly determine investment readiness.

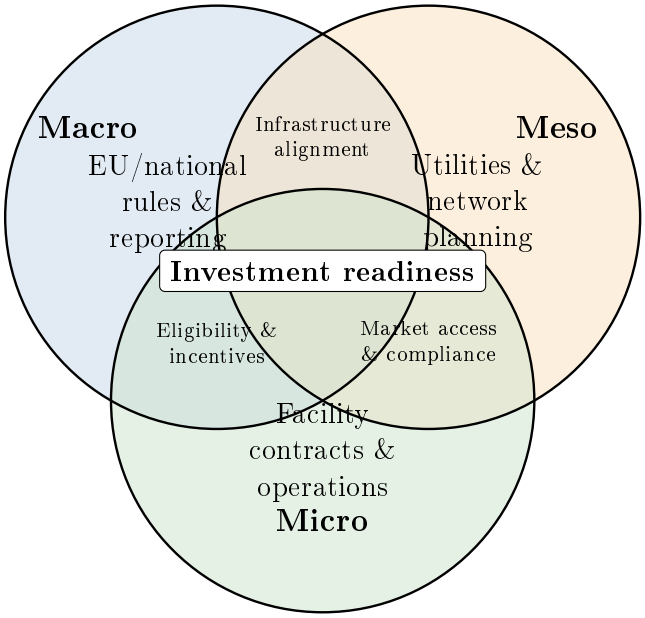


Figure 1: Conceptual macro–meso–micro governance structure. Investment readiness emerges when regulatory signals, infrastructure feasibility, and project-level execution align. This structuring is adapted from the multi-level perspective in socio-technical transitions literature [30] to fit the data-center energy-integration context.

The three circles in Figure 1 represent the macro, meso, and micro levels introduced above. Each pairwise overlap marks a condition that requires two levels to align at once: macro rules must translate into meso infrastructure that can actually deliver them (infrastructure alignment), and meso market access must satisfy the eligibility and incentive structures set at macro level, while micro-level project execution must in turn meet meso market-access and compliance requirements. The region where all three circles overlap, investment readiness, is therefore not a property of any single level; it fails if the macro incentive exists but the meso grid or heat network cannot deliver it, or if both are in place but the micro-level contract structure cannot capture the value. The subsections below examine each level in turn, then return in Section 2.5 to how misalignment across levels stalls otherwise-viable projects.

2.1. Macro level — Policy and market architecture

EU and national policy shape the incentive boundary for data-center energy integration. The European Green Deal and "Fit for 55" package strengthen the value of efficiency, waste-heat reuse, and flexibility [31, 7].

The recast Energy Efficiency Directive introduces reporting obligations for large data centers and codifies KPIs such as PUE, Renewable Energy Factor, and Energy Reuse Factor [9, 29]; voluntary initiatives such as the Climate Neutral Data Centre Pact add sector targets for renewable supply and heat reuse [10]. These metrics increasingly enter due diligence, ESG scoring, and financing conditions, so a metric boundary that ignores system services can distort investment signals.

Digitalization and market-design policy form the second macro channel. The EU action plan for digitalizing the energy system treats flexibility, interoperability, and data availability as prerequisites for renewable integration [8]. Whether data centers can monetize controllable load or batteries depends on balancing-market access, telemetry rules, product design, and congestion-pricing frameworks [32, 12]. Policy and public–private programs can reduce financing risk when they create durable rules or bundled pipelines [33, 34]; unstable incentives increase the option value of waiting.

Carbon pricing adds a third channel: by raising the marginal cost of fossil generation, EU ETS prices increase the relative value of demand reduction, heat reuse, and electrified heat pumps [7]. Stable carbon-price expectations can support long-term heat and flexibility contracts; volatile trajectories raise revenue uncertainty and financing risk [9, 10].

2.2. Meso level — Regional infrastructure and spatial constraint

Meso conditions determine whether macro incentives become feasible projects. Grid capacity is often the binding constraint: large, steady loads can trigger reinforcement, congestion charges, import caps, or even paused connection queues [5, 6]. Projects that credibly reduce peak stress through flexibility, storage, or heat export are therefore better positioned in permitting and connection negotiations.

Land and heat-network access create parallel constraints. Integration equipment expands the footprint, competes with future IT expansion, and can lock projects into sites whose heat demand, renewable resources, and grid access are fixed for the asset life. Heat reuse is technically feasible in many contexts, but its commercial value depends on network proximity, temperature requirements, pipe capacity, tariff treatment, and standardized contracts [11, 35]. These constraints make local infrastructure, not only facility efficiency, central to investment readiness.

Timing is the main meso risk. Data centers can often be built faster than grid reinforcements or district-heating extensions, so service revenues

may be delayed or curtailed after capital is committed. Early alignment with utilities, municipalities, and offtakers is therefore needed to synchronize connection rights, heat-network expansion, and market access. Renewable-resource geography reinforces this logic: Nordic and energy-hub settings can improve PUE and heat-cascading economics, while dense urban clusters may offer strong heat demand but tighter grid and land constraints [36, 37].

2.3. Micro level — Project finance and capital allocation

At project level, investors evaluate integration through NPV, IRR, debt service, and downside risk. Heat exchangers, heat pumps, storage, controllability, and telemetry raise CAPEX but can diversify revenue through heat offtake and flexibility payments. European heat-reuse projects show commercial structures ranging from utility-paid offtake to co-investment and third-party ownership, usually with long-term price agreements [35]; flexibility value depends on market access, telemetry, and performance rules [32, 12].

Uncertainty in energy prices, carbon prices, market access, or connection timing raises the cost of capital. Projects with contracted heat or flexibility revenue can support cheaper debt and higher leverage; speculative service revenues push financing toward higher-return equity. These dynamics encourage staged integration: developers may secure connection rights or heat offtake first, and defer additional storage or controllability until market rules are clearer.

Standard metrics influence this financing logic. Because PUE and related KPIs enter reporting, ESG scoring, and due diligence [9, 10], projects that add integration electricity can look weaker even when they create system value [13, 16]. The model below reflects this tension by comparing service-disabled and service-active regimes.

2.4. Service Channels and Revenue Stacking

Data centers contribute to local energy systems through three service channels: heat recovery, electrical flexibility, and renewable anchoring. Each can improve project economics only if the service is contractible, measurable, and compatible with local infrastructure. Heat export requires calibrated metering and accounting for temperature lift and losses; flexibility requires telemetry and compliance with market performance rules [8, 12].

Waste heat can be recovered from cooling circuits and upgraded through heat exchangers and heat pumps, often from a 25–40°C source stream to district-heating or process-heat levels. Figure 2 summarizes this process,

from recovered heat through upgrading to the commercial interface that determines whether it is investable. Urban waste heat is a material decarbonization resource [38], and the Energy Efficiency Directive adds reporting and feasibility obligations for large facilities [9]. Commercial viability still depends on heat demand, network access, tariffs, curtailment rules, and allocation of temperature-lift responsibility. European project evidence shows utility-paid offtake, co-investment, and third-party ownership models, with long-term contracts central to bankability [11, 35]. Heat prices also interact with carbon prices, since low-carbon heat becomes more valuable as fossil heat is penalized under the same ETS mechanism [7].

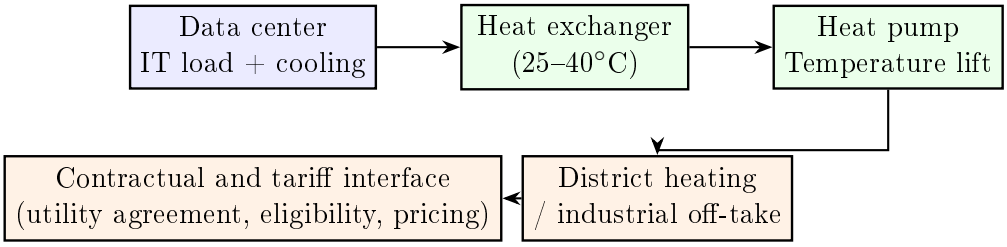


Figure 2: Process flow for recovery and upgrading of data-center waste heat for district heating or industrial off-take, including the contractual and tariff interface that determines commercial viability.

Figure 2 separates the technical steps from the commercial one. The heat exchanger and heat pump only determine whether the recovered stream can be lifted to a usable temperature; they do not determine whether anyone will buy it. That depends on the last box, the contractual and tariff interface, where the utility agreement, network eligibility, and pricing terms are negotiated. A design can clear every technical step in this chain and still fail commercially if the off-take contract is missing, which is why the discussion above treats network proximity, tariff treatment, and contract structure as separate constraints from the heat-recovery hardware itself.

Electrical flexibility comes from controllable IT or cooling loads and battery systems originally installed for reliability. These assets can provide balancing, reserve, or demand-response services when market rules allow participation [12, 18]. The value is both direct, through service revenue, and indirect, through grid-access and congestion-risk mitigation. Product definitions, telemetry, settlement intervals, and opportunity costs determine whether this value is investable [19].

Renewable anchoring combines procurement with physical siting. Corporate power purchase agreements can reduce exposure to wholesale-price volatility, while energy-hub concepts co-locate data centers with renewables, storage, and flexible loads [37]. Nordic demonstrations show how clean generation and cold climates can support lower overhead and heat cascading [36]. These benefits remain place-specific because land, grid access, and heat-network proximity constrain where they can be realized [39].

2.5. Cross-Level Misalignment and the KPI Gap

Integration is often slowed by cross-level misalignment: policy encourages decarbonization and reporting, local infrastructure requires grid relief and spatial coordination, and project finance depends on predictable cash flows. PUE remains the dominant reporting metric [40, 13], but it classifies all non-IT electricity as overhead. Heat pumps, storage, and controllable cooling can therefore worsen PUE even when they create external value, a limitation also noted in empirical and review work [16, 15].

Because PUE and related indicators are now used in regulation, ESG screening, and financing [9, 29], this boundary matters for investment. A service-active project can improve system outcomes yet appear weaker under conventional reporting. The siPUE metric introduced below is designed as a translation layer: it preserves the PUE denominator and numerator logic, but credits monetized local-system benefits separately from private revenues and costs.

The model reflects the same context. Demand follows IT workload profiles; delay enters through lifecycle availability; electricity-side constraints use hourly balance and a three-node DC-OPF; heat, flexibility, storage, carbon, land, and lease costs enter through explicit physical and accounting ledgers.

3. Methodological Contribution

This section defines siPUE and the optimization framework used to test it. siPUE keeps the PUE structure but translates the independent system-cost benefit into an energy-equivalent numerator credit. The benefit includes avoided resource, transmission, peak, and carbon-damage cost; private heat and flexibility revenue remain in the market-cost objective.

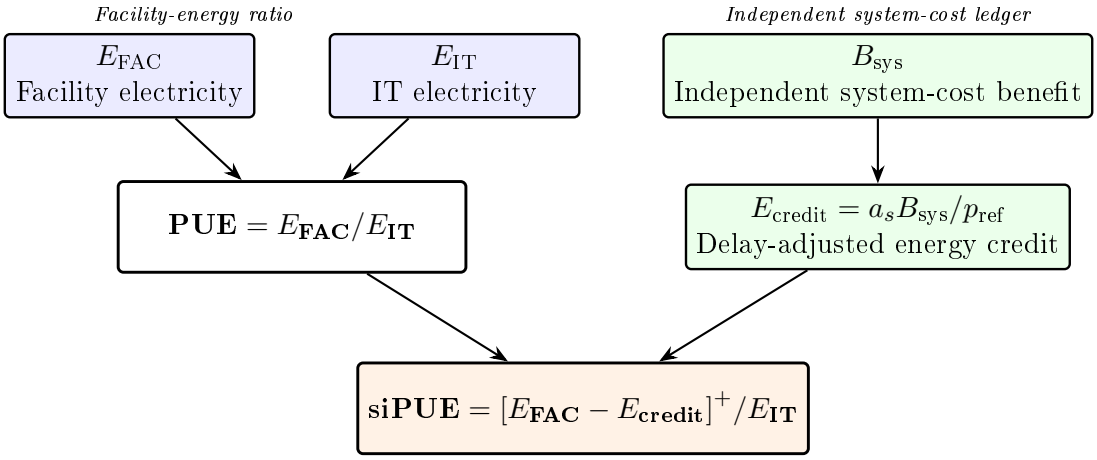
The model evaluates capital decisions (heat exchangers, heat pumps, flexibility capacity, batteries, land, and the locked 15% efficiency retrofit) together with hourly operation. Lifecycle market cost includes electricity,

carbon-market payments, capex, land, congestion, service revenues, nodal import constraints, and construction-delay effects.

The main counterfactual is the *services-disabled* comparator: services are disabled, but the 15% efficiency retrofit and physical constraints are kept the same as in the service-active cases. An identity check with no active services is retained only to verify metric behavior. The *baseline* case is a service-active design; the *robust* case is one min–max-regret design per archetype. Baseline, robust, and scenario-perfect cases share the same carbon cap as the services-disabled comparator, applied per scenario as

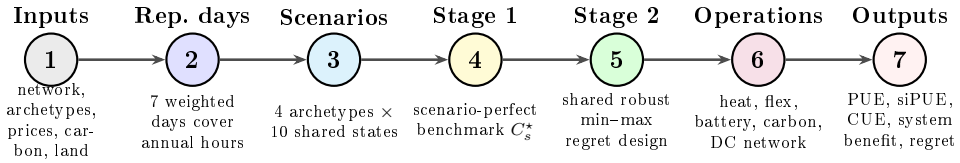
$$\mathrm{Cap}_s^{\mathrm{applied}} = \max\left(\mathrm{Cap}_s^{\mathrm{policy}},\ E_{s,\min}^{\text{services-disabled}}\right), \tag{1}$$

where $\mathrm{Cap}_s^{\mathrm{policy}}$ targets a 5% reduction in attributable emissions relative to the services-disabled comparator's own unconstrained, cost-minimizing emissions in scenario s (not a zero-efficiency design), and $E_{s,\min}^{\text{services-disabled}}$ is that same comparator's true minimum achievable emissions, found by minimizing emissions directly rather than cost. The max only guards against scenarios where the 5% target is physically unreachable for that design; in the canonical rows the applied cap exceeds the policy target by at most 0.10% (mean 0.05%), so the enforced cap is within a tenth of a percentage point of the stated 5% reduction in every row, and the floor is never the dominant constraint.



No active services $\Rightarrow B_{sys} = 0 \Rightarrow E_{credit} = 0 \Rightarrow siPUE = PUE$

Figure 3: Metric logic for siPUE. PUE (left branch) is the internal facility-energy ratio, computed only from measured electricity. siPUE (bottom) subtracts a delay-adjusted, energy-equivalent credit for the independent system-cost benefit (right branch) from the PUE numerator. When no services are active, the credit is zero and siPUE equals PUE.



The process separates scenario design, optimization, operation, and KPI accounting so policy and business claims remain tied to distinct model steps.

Figure 4: Model pipeline from inputs and scenarios to robust design and KPI outputs.

3.1. Definition and interpretation of siPUE

PUE is defined as

$$\mathrm{PUE} = \frac{E_{\mathrm{FAC}}}{E_{\mathrm{IT}}}, \tag{2}$$

where E_{FAC} is modeled facility electricity and E_{IT} is delivered IT electricity. siPUE keeps this denominator and adjusts the numerator by a system-benefit credit. Let superscript 0 denote the services-disabled comparator and d a service-active design. The independent system-cost ledger is

$$C^d_{\mathrm{ind}} = C^d_{\mathrm{el}} + C^d_{\mathrm{trans}} + C^d_{\mathrm{peak}} + C^d_{\mathrm{CO2,dam}} - V^d_{\mathrm{heat}} - V^d_{\mathrm{flex}}, \tag{3}$$

where C_{el}, C_{trans}, C_{peak}, and $C_{\mathrm{CO2,dam}}$ are electricity-resource, network, peak-capacity, and carbon-damage costs, and V_{heat} and V_{flex} are avoided heat and flexibility costs. V_{heat} is avoided boiler fuel cost plus avoided variable O&M only; it excludes any carbon or emissions term. V_{flex} is a fixed avoided reserve-capacity value, independent of carbon pricing. The carbon-damage value of the emissions avoided by displacing boiler heat or thermal generation is credited separately and additively through $C_{\mathrm{CO2,dam}}$, which is computed from the same physical emissions ledger used for V_{heat}'s displaced fuel but from a distinct, non-overlapping cost component (social carbon cost times avoided tCO2, not fuel price), so no term is counted in both places. The credited system benefit is

$$B^{d}_{\mathrm{sys}} = C^{0}_{\mathrm{ind}} - C^{d}_{\mathrm{ind}}. \tag{4}$$

Positive terms in B_{sys} are avoided independent system costs; negative terms are additional system burdens. Components are aggregated before any clipping, so a design with negative net system benefit receives a negative credit and siPUE rises relative to PUE.

Table 2: Accounting boundary for the siPUE system-benefit term.

Included in B_{sys}	Excluded from B_{sys}
Electricity resource-cost changes, transmission or network-cost changes, independent peak-capacity effects, carbon-damage changes, avoided heat-system cost, and avoided flexibility-system cost.	Private heat and flexibility revenues, IT service revenue, ETS or carbon-market payments, capex, land lease or opportunity cost, private demand charges, taxes, subsidies, and financing terms.

This boundary keeps private settlement and public system value separate: heat and flexibility revenues enter the market objective, ETS payments stay in the market ledger, and carbon damage enters only the independent system ledger.

Table 3: Dimensional and temporal basis of the siPUE terms.

Term	Unit	Temporal basis	Discounted?
E_{FAC}	MWh/year	Representative modeled operating year.	No
E_{IT}	MWh/year	Same representative modeled operating year.	No
B_{sys}	EUR/year	Annual system-benefit difference between the services-disabled comparator and design d.	No
p_{ref}	EUR/MWh	Frozen base-case reference price used only for unit conversion.	No
a_s	dimensionless	Lifecycle availability factor applied once to represent delayed useful benefit.	No

The delay-adjusted energy credit is

$$E_{\mathrm{credit}} = a_s \frac{B_{\mathrm{sys}}}{p_{\mathrm{ref}}}, \tag{5}$$

where p_{ref} is a fixed unit-conversion price and $a_s \in [0, 1]$ is the lifecycle availability factor after construction delay. In the canonical run, $p_{\mathrm{ref}} = 56.63$ EUR/MWh, based on the electricity-resource marginal cost at the data-center node in the *identity no-services* case, not the services-disabled comparator used as the reference case elsewhere. The identity case is used here because it has zero efficiency investment and zero service capacity, so its marginal price cannot itself be shifted by the design choices siPUE evaluates; anchoring p_{ref} to the services-disabled comparator instead would let its locked efficiency retrofit and carbon-cap treatment influence the unit-conversion factor common to every case. ETS payments and carbon damage are excluded because they are handled in separate ledgers. Freezing p_{ref} at this design-independent value prevents the metric from improving through the conversion factor itself.

siPUE is

$$\mathrm{siPUE} = \frac{[E_{\mathrm{FAC}} - E_{\mathrm{credit}}]^+}{E_{\mathrm{IT}}}, \tag{6}$$

Table 4: Reference-price sensitivity for the canonical robust rows. Dispatch and ledgers are held fixed; only the value-to-energy conversion factor changes. Average siPUE uses IT-energy weights across archetypes.

Reference-price case	Interpretation	Average siPUE
$0.5p_{\mathrm{ref}}$	Low conversion price; stronger energy credit	0.961
$1.0p_{\mathrm{ref}}$	Canonical frozen reference price	1.152
$1.5p_{\mathrm{ref}}$	High conversion price; weaker energy credit	1.216

with $[z]^+ = \max(0, z)$. Internal property checks verify nonnegativity, reduction to PUE when no services are active, delay behavior, carbon-ledger separation, scenario count, scale normalization, benefit monotonicity, and reference-price monotonicity.

3.2. Reference-price sensitivity and anti-gaming rule

The reference price is the main conversion choice in siPUE. Holding design and operation fixed, a multiplier k gives

$$\mathrm{siPUE}(k) = \frac{[E_{\mathrm{FAC}} - a_s B_{\mathrm{sys}}/(k p_{\mathrm{ref}})]^+}{E_{\mathrm{IT}}}. \tag{7}$$

The anti-gaming rule is to set p_{ref} ex ante and hold it constant across designs and uncertainty states. Table 4 changes only Eq. 7; dispatch, heat delivery, emissions, PUE, and B_{sys} are unchanged.

The sensitivity shows that p_{ref} affects the magnitude of the credit, even when all physical and monetary ledgers are unchanged. siPUE should therefore be reported with the reference-price rule and a sensitivity band. It is a companion KPI, not a thermodynamic-efficiency measure.

3.3. Two-stage design model

Let $s \in \mathcal{S}$ index the shared uncertainty states, x denote design variables, and y_s denote hourly operational variables. Stage 1 solves a scenario-perfect benchmark:

$$C_s^\star = \min_{x, y_s} C_s(x, y_s), \tag{8}$$

where C_s is lifecycle market cost. Stage 2 chooses one shared design $\hat{x}$ per archetype using minimax regret [25, 26]. The ten states are unweighted, and each archetype is solved independently. Regret is normalized by each scenario's present-value-discounted lifetime IT energy $E_{\mathrm{IT},s}^{\mathrm{PV}}$:

$$R_s(\hat{x}) = \frac{C_s(\hat{x}, y_s) - C_s^\star}{\max(\epsilon, E_{\mathrm{IT},s}^{\mathrm{PV}})}. \tag{9}$$

The robust design is selected in two passes: minimize worst normalized regret, then fix $Z \leq Z^\star(1+\tau)$ and minimize mean lifecycle cost within that band. This lexicographic tie-break [27, 28] uses $\tau = 10^{-7}$ in the canonical run. The publication-facing cases are the services-disabled comparator, baseline service-active design, robust shared design, and scenario-perfect benchmark; the no-active-services case is used only to verify metric identity.

The canonical run sets the efficiency reduction to 0.15 in the services-disabled comparator, baseline, robust, and scenario-perfect cases. Optimized capacities are heat exchanger, heat pump, flexibility, battery energy and power, and associated land use. The identity check keeps service capacity and efficiency reduction at zero. This locked retrofit is not an assumption that favors the service-active designs: re-solving the services-disabled design in every one of the 40 canonical rows with efficiency chosen freely from the same discrete candidate set (0% or 15%) selects 0.15 in all 40 rows, so the services-disabled comparator's own unconstrained economics already favor the retrofit independently of any service.

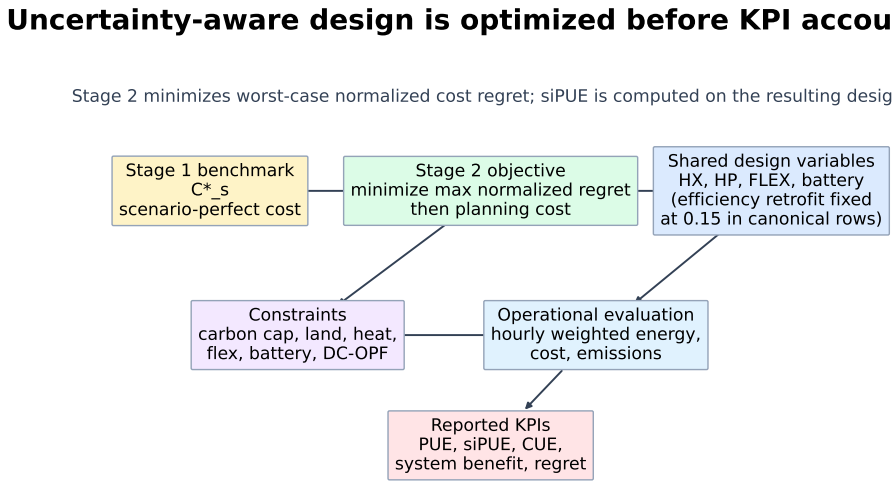


Figure 5: Optimization map linking design variables, scenario-perfect benchmarks, normalized regret, and physical constraints.

3.4. Operational constraints and ledgers

Hourly operation is a linear program. The facility balance links grid import, battery operation, IT load, cooling, and heat-pump electricity. Heat export is bounded by heat-exchanger and heat-pump capacity, recoverable heat, pipe limits, and district-heating acceptance. Flexibility is modeled as reserve availability, with routine activation outside the settlement basis [18, 19]. Deliverability is enforced through IT-load headroom, controllable capacity, unused battery inverter capacity, stored energy above the UPS reserve, and parasitic overhead.

The electricity system is a stylized three-node DC-OPF with thermal generation, renewable availability, line capacities, and nodal settlement prices. Carbon costs follow the market/independent ledger split in Table 2. Land is represented as lease or opportunity cost tied to core facility, cooling, heat, flexibility, and battery footprints.

Network structure is illustrated in Fig. 6 and Fig. 7. These schematics correspond to the nodal representations used in the optimization model: a 3-node DC-OPF for the electricity system and a simplified heat-recovery network with storage.

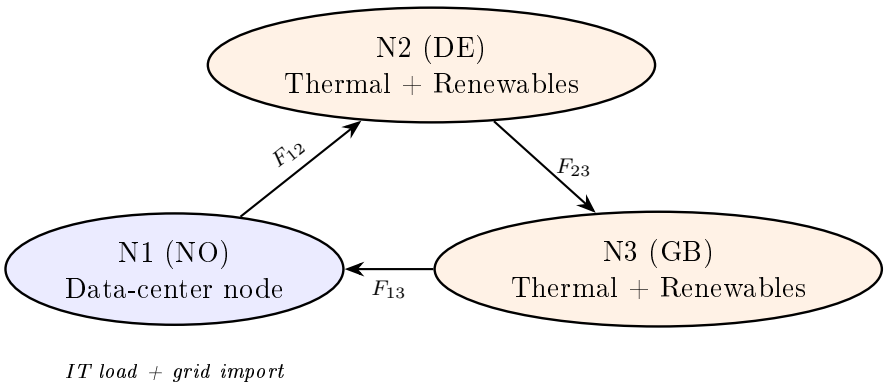


Figure 6: Stylized three-node DC-OPF electricity network. Each node represents one modeled country group: N1 (Nordics – Norway, Sweden, Denmark, Finland, Iceland), N2 (Central Europe – Germany, Belgium, Netherlands, France), and N3 (United Kingdom); one representative country is shown per node. F_{12}, F_{23}, and F_{13} are the linearized DC power flows between node pairs; arrow direction is illustrative, since flows are signed variables that can reverse hour to hour. The data-center node shown here (N1) is illustrative – each archetype's own scenario location determines which node it actually attaches to.

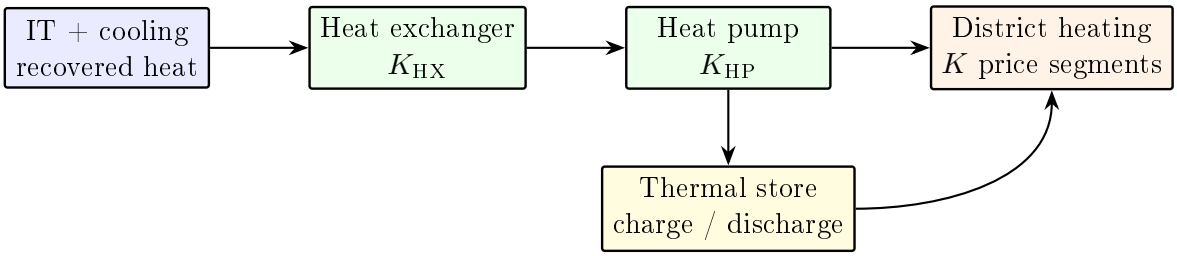


Figure 7: Simplified heat-recovery network with storage. Recovered heat from IT and cooling loads passes through the heat exchanger and heat pump to become delivered heat, is optionally buffered in a thermal store, and is sold into up to K district-heating price segments.

3.5. Data and scenario design

The model uses a calibrated 8,760-hour, three-node time series compressed into seven representative days. Stratified medoid clustering selects representative days from seasonal strata using standardized fuel-price, load, renewable, net-load, calendar, ambient-temperature, and district-heating features. Cluster weights preserve annual hours. Fixed annual terms enter once outside the representative-day weighting. The seven days are concatenated

into one operating horizon, so battery state of charge and peak import are evaluated across the combined sample rather than reset daily.

The canonical scenario set has four archetypes – AI/HPC, hyperscale cloud, colocation, and enterprise/edge – each facing ten shared uncertainty states, giving 40 rows. States vary service prices, heat conditions, construction delay, carbon caps, and operating conditions. Archetype base recoverable-heat fractions are 0.20–0.50 [11, 41]; state multipliers raise effective LP inputs to 0.17–0.575 in the canonical rows. Outputs are normalized by IT energy or IT capacity where appropriate.

Table 5: Canonical scenario structure used in the model evaluation.

Dimension	Count	Role in the model
Archetypes	4	AI/HPC, hyperscale cloud, colocation, enterprise/edge
Shared uncertainty states	10	Heat, flexibility, carbon-cap, delay, and service conditions
Canonical rows	40	One row for each archetype and uncertainty-state pair
Publication-facing cases	3 (+1 benchmark)	services-disabled comparator, baseline service-active, robust shared design; plus the scenario-perfect benchmark reported alongside them
Representative days	7	Weighted annual operating approximation

Table 6 and Table 7 report the ten shared uncertainty states in full; the scenario generator and seed are released with the code (Code availability, below).

Table 6: Shared uncertainty states: absolute input levels. El. price and heat price are EUR/MWh(th); flex price is EUR/MW-h; ETS and carbon-damage prices are EUR/tCO2; delay is months.

State	Loc.	El. price	El. vol.	Heat price	Heat dem.	Flex price	ETS	Damage	Delay
1 Central-base	Netherlands	65	18	58	50	18	65	120	6
2 Clean-cool	Norway	48	13	48	62	16	45	80	0
3 High-ETS	Germany	72	24	62	48	21	105	160	6
4 Heat-rich	Denmark	58	20	78	78	20	75	140	0
5 Flex-rich	UK	82	30	52	30	31	80	130	12
6 Cooling-stress	France	70	22	55	38	18	70	110	6
7 Congestion-stress	Belgium	76	27	60	44	24	85	180	6
8 Storage-limited	Sweden	54	19	50	55	17	60	100	0
9 Cost-stress	Netherlands	74	25	56	42	19	90	200	12
10 Service-upside	Denmark	60	21	82	70	29	75	150	6

Table 7: Shared uncertainty states: multipliers applied to archetype base values. Cong. = congestion, Cool. OH = cooling overhead, Free cool. = free-cooling availability, Heat rec. = heat-recovery fraction, HP COP = heat-pump coefficient of performance, Flex avail. = flexibility availability, Storage = storage availability, Capex and Land are cost multipliers.

State	Cong.	Cool. OH	Free cool.	Heat rec.	HP COP	Flex avail.	Storage	Capex	Land
1 Central-base	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
2 Clean-cool	1.20	0.82	1.25	1.10	1.08	0.95	1.10	0.90	0.80
3 High-ETS	0.90	1.05	0.95	0.95	0.95	0.90	0.90	1.00	1.10
4 Heat-rich	0.85	0.92	1.15	1.15	1.05	1.00	1.00	1.05	0.95
5 Flex-rich	0.75	1.08	0.90	0.90	0.90	1.20	1.00	1.00	1.05
6 Cooling-stress	1.05	1.28	0.70	0.85	0.92	0.90	0.90	1.10	1.00
7 Congestion-stress	0.60	1.12	0.90	1.00	0.95	1.05	0.90	1.05	1.10
8 Storage-limited	1.10	0.88	1.20	1.05	1.05	0.95	0.35	1.00	0.90
9 Cost-stress	0.80	1.15	0.85	0.90	0.90	0.90	0.80	1.35	1.30
10 Service-upside	0.95	0.95	1.10	1.15	1.08	1.15	1.15	0.90	0.95

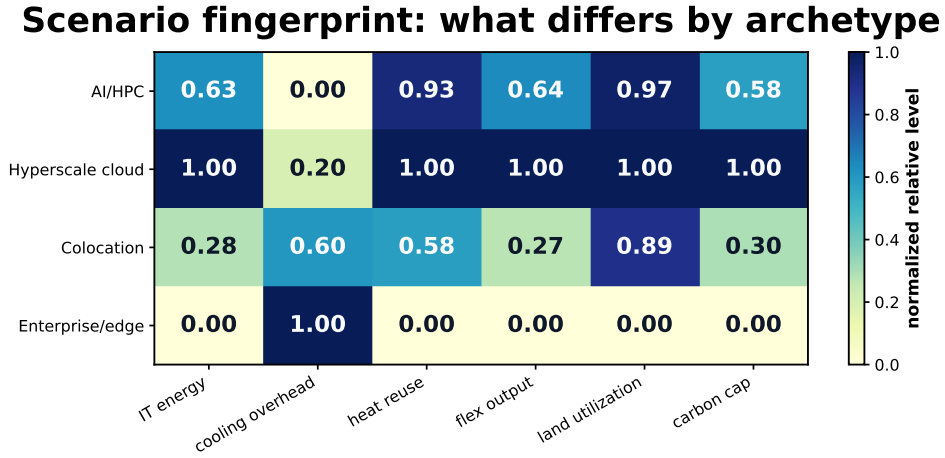
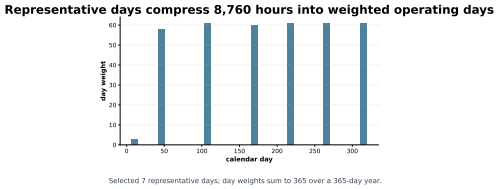
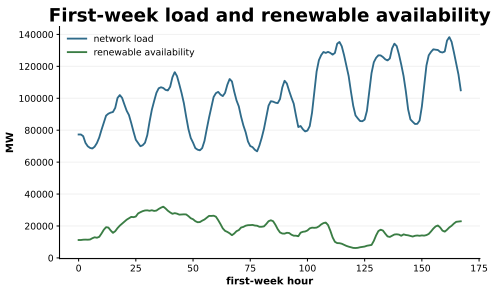


Figure 8: Input scenario fingerprint, normalized within each column to show relative scenario structure.

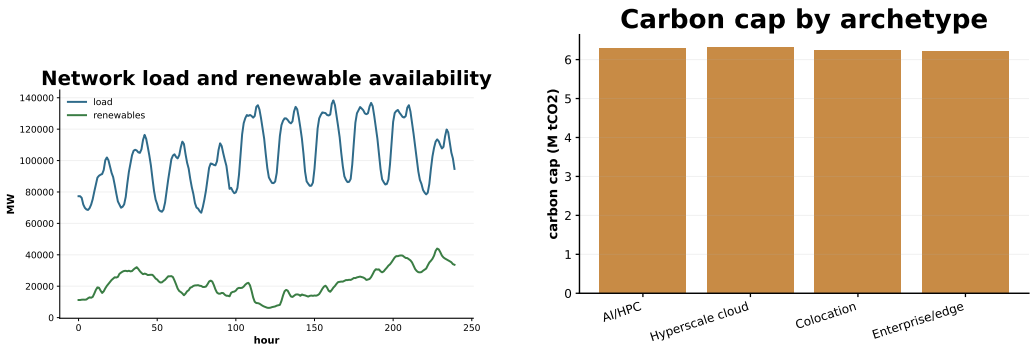


(a) Selected representative days and weights.



(b) First-week load and renewable profiles.

Figure 9: Representative-day compression and hourly system context.



(a) Load and renewables.

(b) Carbon-cap context.

Figure 10: Load, renewable, and carbon-cap context used by the operational model. Network topology is shown separately in Fig. 6.

4. Results and Discussion

The results first report the PUE–siPUE split, then qualify it through decomposition, compact robustness checks, sensitivity, and face-validity comparison [42, 43].

4.1. PUE and siPUE move differently

Table 8 reports canonical averages weighted by each archetype's annual IT energy. The services-disabled comparator, with the same efficiency retrofit and carbon-cap treatment as the service-active cases, gives PUE and siPUE values of 1.167 and is the reference case.

Against that comparator, the baseline service-active design raises average PUE to 1.238 (+0.071) while lowering average siPUE to 1.139 (−0.028), earning 5.603 EUR/MWh IT of delay-adjusted system benefit – about 1.40 times its facility-energy burden. The robust design raises PUE further, to 1.343 (+0.176), and lowers siPUE to 1.152 (−0.015), earning a larger absolute benefit, 10.825 EUR/MWh IT, but a smaller credit-to-burden ratio of about 1.08. Robust is therefore not simply "better according to siPUE" than baseline – baseline has the lower average siPUE – but it creates more absolute system value at a lower benefit-to-burden ratio, consistent with diminishing marginal system value as integration intensity scales. The resulting net siPUE movement is modest because credit and burden are close in size (ratios of 1.40 and 1.08), not because the underlying heat, flexibility, and carbon-damage prices are too small to move the metric: with the same

prices, a design chosen to minimize siPUE regret directly rather than cost regret lowers siPUE by roughly seven times as much (Section 4.5). Baseline and robust are reported because cost-regret robustness is the paper's stated design criterion, and their modest siPUE movement is an honest consequence of that choice, not a ceiling on what the priced services can achieve.

The credit is an economic conversion, not a physical energy saving. The carbon-exclusion check below shows that the average siPUE improvement disappears once carbon damage is removed from the ledger.

Table 8: Compact metric benchmark across the 40 canonical rows, weighted by each archetype's annual IT energy. Carbon, heat, and emissions are kept as separate ledgers with their own labels.

Design case	PUE	Delay-adj. siPUE	Post-constr. siPUE	System benefit (EUR/MWh IT)	Benefit excl. carbon (EUR/MWh IT)
Services-disabled comparator	1.167	1.167	1.167	0.000	0.000
Baseline	1.238	1.139	1.135	5.603	3.433
Robust	1.343	1.152	1.145	10.825	5.161

Baseline has lower average siPUE, but robust has higher heat-reuse intensity and more absolute system value because it is selected for cost-regret robustness, not minimum siPUE.

At the row level, siPUE is lower than the services-disabled comparator in 36 of 40 baseline rows and 23 of 40 robust rows; PUE is higher in all baseline and robust rows. Figure 11 shows the paired baseline-to-robust movement relative to the services-disabled comparator.

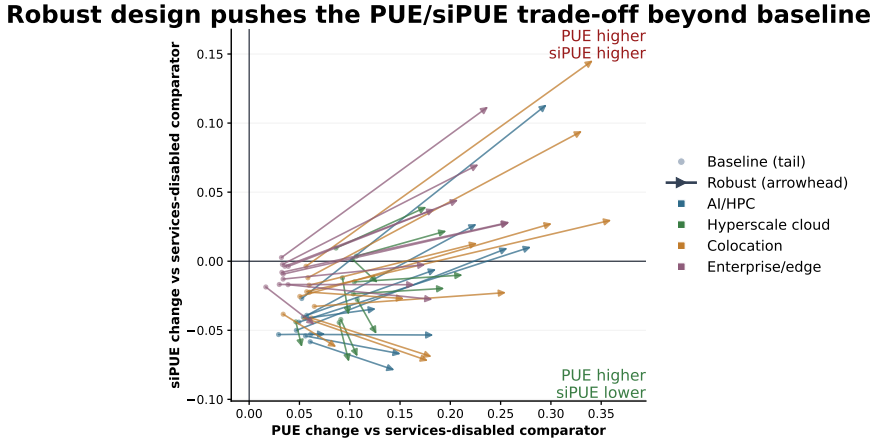


Figure 11: PUE and siPUE tradeoff compass. Each arrow moves one scenario from the baseline design to the robust design, with both metrics measured as change relative to the constrained services-disabled comparator.

4.2. Credit decomposition

Table 9 decomposes the robust siPUE credit. Heat, flexibility, and carbon-damage benefits are partly offset by electricity-resource, peak-capacity, and transmission burdens; construction delay then weakens the net credit.

The ablation shows why PUE rises while siPUE can fall: the design uses more facility electricity, but creates system value in the independent ledger. Carbon damage is material. Excluding it raises IT-weighted robust siPUE from 1.152 to 1.252, above the fair comparator's 1.167, and only 4 of 40 robust rows remain below the comparator. The result is therefore conditional on the stated social-carbon valuation.

A single full-versus-zero-carbon comparison only bounds this dependence; the break-even social cost of carbon,

$$\mathrm{SCC}^{\mathrm{BE}} = \frac{(\mathrm{PUE} - \mathrm{siPUE}_{\mathrm{fair}})\, p_{\mathrm{ref}} - V_{\text{non-carbon}}}{E_{\text{avoided,tCO2}}}, \tag{10}$$

is more informative, where $V_{\text{non-carbon}}$ is the delay-adjusted heat-plus-flexibility credit and $E_{\text{avoided,tCO2}}$ is avoided emissions per MWh IT. At the canonical SCC = 120 EUR/tCO2, the IT-weighted robust break-even value is $\mathrm{SCC}^{\mathrm{BE}} \approx$ 102 EUR/tCO2 – below the canonical value, so the reported improvement

Table 9: Component ablation of the robust siPUE credit across the 40 canonical rows, weighted by each archetype's annual IT energy. Values are EUR/MWh IT. Components above the delay row are before lifecycle delay; the final row is the delay-adjusted benefit credited by siPUE.

Component	Contribution	Interpretation
Heat system value	10.953	Avoided independent heat-system cost
Flexibility system value	3.799	Avoided independent flexibility cost
Carbon damage	5.864	Avoided carbon-damage ledger cost
Peak capacity	-1.182	Additional independent peak-capacity burden
Electricity resource cost	-7.834	Additional power-system resource burden
Transmission / network cost	-0.371	Additional network-flow burden
Undelayed system benefit	11.231	Net benefit before delay adjustment
Construction-delay adjustment	-0.405	Lost credit from delayed availability
Delay-adjusted system benefit	10.825	Benefit entering the siPUE numerator

holds with some margin, but well inside the 80–250 EUR/tCO2 policy-appraisal range used to calibrate the canonical assumption. By archetype, SCC^{BE} is approximately 106 EUR/tCO2 for AI/HPC, 77 for Hyperscale cloud, 125 for Colocation, and 144 for Enterprise/edge. Only AI/HPC and Hyperscale cloud have a break-even value below the canonical 120 EUR/tCO2, so at the canonical valuation their archetype-average robust siPUE is already below the fair comparator; Colocation and Enterprise/edge have a break-even value above 120 EUR/tCO2, so their archetype-average robust siPUE has not yet crossed the fair comparator at the canonical valuation. The IT-weighted overall improvement therefore comes disproportionately from AI/HPC and Hyperscale cloud, not uniformly from all four archetypes.

4.3. Design portfolios and system value

The baseline case is an assumed incumbent portfolio, not an independent optimization: a per-MW-IT intensity for heat-exchanger, flexibility, and battery capacity is scaled by an archetype-specific multiplier reflecting typical industry adoption (hyperscale campuses assumed to have invested most in heat recovery and flexibility; enterprise/edge assumed to have invested least), with heat-pump capacity sized to match via the COP-dependent source-to-power ratio and the same locked 15% efficiency retrofit used elsewhere. Table 10 reports the resulting capacities.

Table 10: Baseline (incumbent) design capacities by archetype, an assumed portfolio scaled from a per-MW-IT intensity by an archetype-specific adoption multiplier (Section 3), not an independent optimization.

Archetype	HX (MWth)	HP (MW)	Flex (MW)	Batt. E (MWh)	Batt. P (MW)	Eff.
AI/HPC	5.50	1.96	4.80	22.40	5.60	0.15
Hyperscale cloud	15.00	6.00	4.80	48.00	12.00	0.15
Colocation	2.50	1.19	1.10	12.00	3.00	0.15
Enterprise/edge	0.40	0.22	0.19	2.24	0.56	0.15

Figure 12 shows robust technology intensities by archetype, and Figure 13 decomposes the resulting system value.

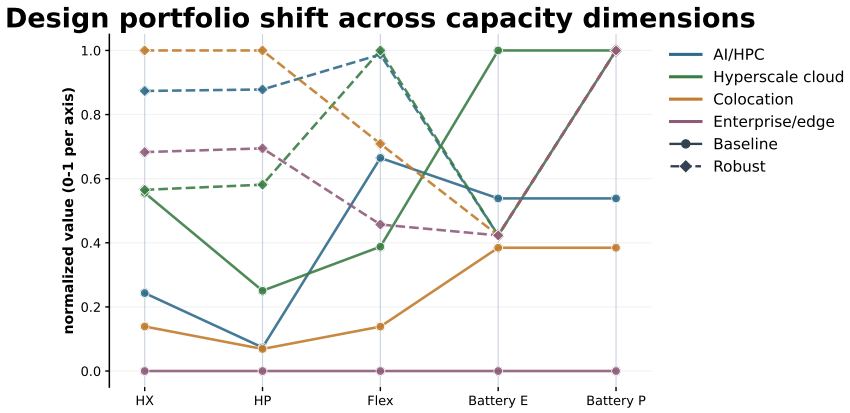


Figure 12: Baseline and robust design portfolios by archetype, normalized per MW IT. Each capacity axis is independently normalized to make portfolio shape comparable across archetypes; it does not preserve absolute scale. Table 11 reports the underlying absolute capacities.

Table 11: Robust design capacities by archetype in absolute units, underlying the per-MW-IT-normalized portfolio shapes in Figure 12.

Archetype	HX (MWth)	HP (MW)	Flex (MW)	Batt. E (MWh)	Batt. P (MW)
AI/HPC	14.57	11.54	6.66	20.0	8.0
Hyperscale cloud	15.20	12.03	10.10	30.0	12.0
Colocation	10.24	8.11	3.16	12.5	5.0
Enterprise/edge	2.37	1.87	0.72	4.0	1.6

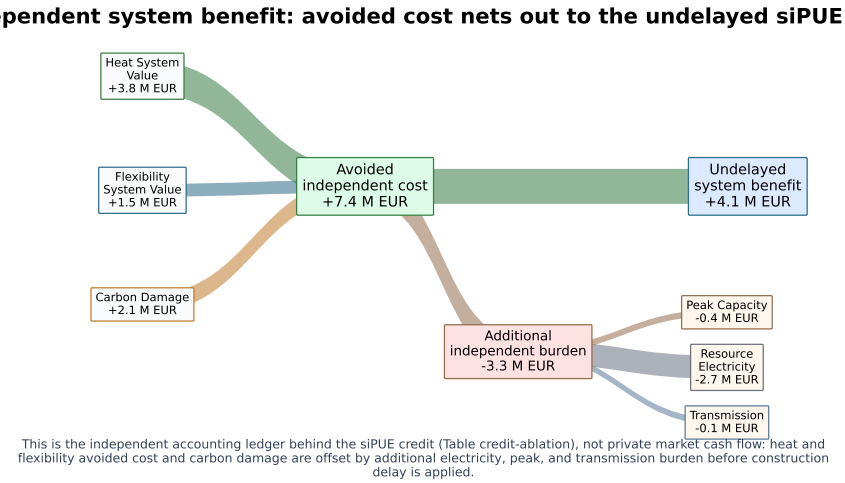


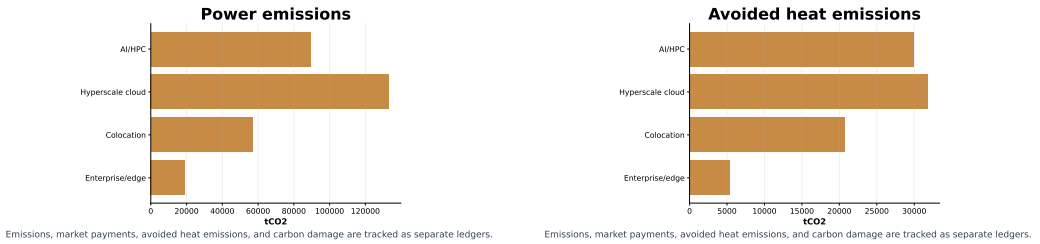
Figure 13: Independent system-benefit ledger for the robust design (the accounting that defines the siPUE credit): avoided heat and flexibility cost and carbon damage are reported separately from the additional electricity, peak, and transmission burden. This is not a market cash-flow diagram – private revenue, capex, and land cost are excluded (Table 2).

4.4. Carbon and delay accounting

Carbon-damage valuation can change monetized system benefit without changing physical emissions, so it is retained as a separate ledger item (Table 12) and excluded from the main physical sensitivity factors below.

Table 12: Carbon mechanisms and accounting treatment.

Mechanism	Purpose	In optimization?	In siPUE credit?
ETS payment	Private compliance and dispatch cost in the market ledger.	Yes	No
Carbon-damage value	External damage value in the independent accounting ledger.	Ledger only	Yes
Physical carbon cap	Row-level planning constraint shared by the services-disabled comparator, baseline, robust, and scenario-perfect cases.	Yes	No direct monetary credit



(a) Power emissions.

(b) Avoided heat emissions.

Figure 14: Carbon-ledger separation in the robust design.

Construction delay affects siPUE through the availability factor in Eq. 5. It reduces the credit for useful system benefit while leaving the underlying facility energy accounting intact. The canonical outputs distinguish post-construction siPUE from delay-adjusted siPUE for this reason.

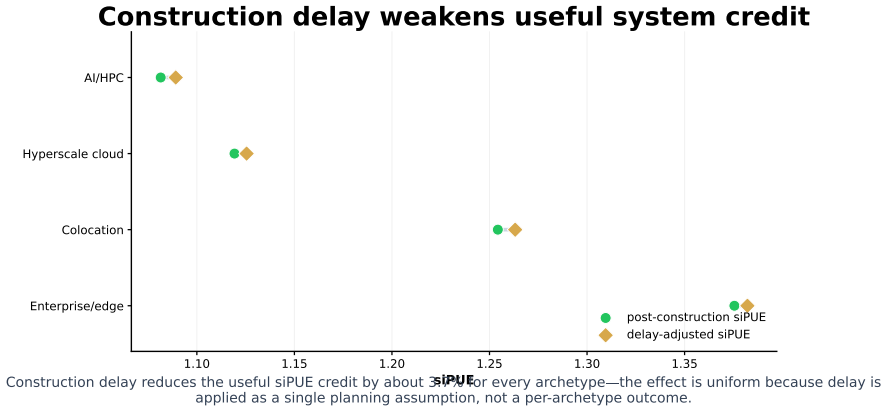


Figure 15: Bridge from post-construction siPUE to delay-adjusted siPUE. Delay weakens the useful system credit before it enters the metric.

4.5. Robustness checks

These checks support internal consistency, not external validation. They confirm that siPUE remains finite and nonnegative, reduces to PUE when no services are active, responds monotonically to benefit and delay inputs, and keeps the carbon ledger separate. Held-out and synthetic out-of-sample runs remain feasible. A direct siPUE-regret re-solve also shows that cost-regret robustness is not the same as siPUE minimization.

Table 13: Compact internal robustness checks.

Check	Result
Metric properties	Nonnegativity, PUE identity with no active services, delay behavior, benefit monotonicity, scale normalization, and carbon-ledger separation all pass.
Held-out and out-of-sample operation	12/12 held-out rows and 40/40 synthetic out-of-sample rows remain feasible; mean cost regret is 1.065 and 0.402 EUR/MWh IT, respectively.
Direct siPUE-regret alternative	Optimizing siPUE regret directly lowers weighted mean siPUE from 1.152 to 1.067 in-sample and from 1.179 to 1.088 out-of-sample; average lifecycle-cost difference remains below 0.11%.

4.6. Sensitivity analysis

The manuscript sensitivity varies five physical or delivery factors: heat price, flexibility price, cooling overhead, land availability, and construction delay. Carbon damage remains a separate valuation ledger.

Table 14: All-archetype sensitivity results for the robust design across the 40-row scenario set for each sensitivity case, weighted by each archetype's annual IT energy. The land availability "unconstrained" row uses the same design as "high" (PUE, siPUE, and benefit are unchanged) but divides the same absolute land use by an arbitrarily large nominal reference limit rather than a physical constraint, so its land-use percentage is not a meaningful utilization ratio and is reported only for table-format consistency.

Factor	Level	PUE	siPUE	Benefit (EUR/MWh IT)	Land use (%)
Canonical	base	1.343	1.152	10.825	75.2
Heat price	low	1.239	1.117	6.895	60.4
Heat price	high	1.365	1.161	11.554	79.9
Flexibility price	low	1.344	1.171	9.840	73.1
Flexibility price	high	1.342	1.150	10.907	75.8
Cooling overhead	low	1.308	1.118	10.750	74.4
Cooling overhead	high	1.378	1.186	10.898	75.9
Land availability	low	1.316	1.163	8.696	96.5
Land availability	high	1.343	1.152	10.825	62.6
Land availability	unconstrained	1.343	1.152	10.825	0.8
Construction delay	0 months	1.347	1.146	11.403	76.9
Construction delay	12 months	1.343	1.161	10.279	74.9

Each sensitivity level re-optimizes the shared design and its dispatch under the perturbed factor; it is not a one-factor accounting perturbation with the canonical design and dispatch held fixed. This is why the low heat-price row shows both a lower system benefit (6.895 versus 10.825 EUR/MWh IT) and a lower siPUE (1.117 versus 1.152) than the canonical case: a low heat price makes heat recovery less attractive, so the re-optimized design shifts toward a smaller, less electricity-intensive portfolio, lowering PUE (1.239 versus 1.343) enough to lower siPUE too even though the credit itself shrinks. The

sensitivity therefore combines design adaptation and metric response, and should not be read as showing that lower heat prices improve siPUE holding the design fixed. The sensitivity analysis separates the facility-energy burden from the value credit. Cooling overhead changes both PUE and siPUE. Flexibility-price shifts mostly affect siPUE. Delay weakens the credit. Low land availability binds in three robust rows and reduces system benefit.

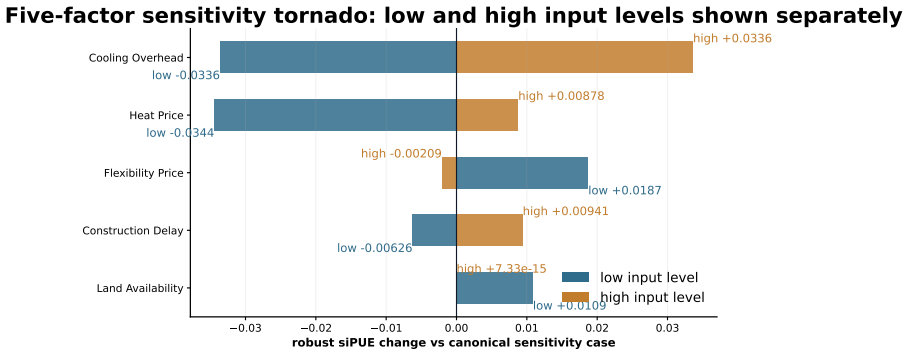


Figure 16: Five-factor sensitivity tornado for robust siPUE. Each factor's low and high input levels are shown as separate bars relative to the canonical value (0), not as a single low-to-high range, so the direction of each level's effect is explicit. Each level re-optimizes design and dispatch rather than perturbing the canonical design's operation alone (Section 4). Carbon damage is excluded from the sensitivity factors.

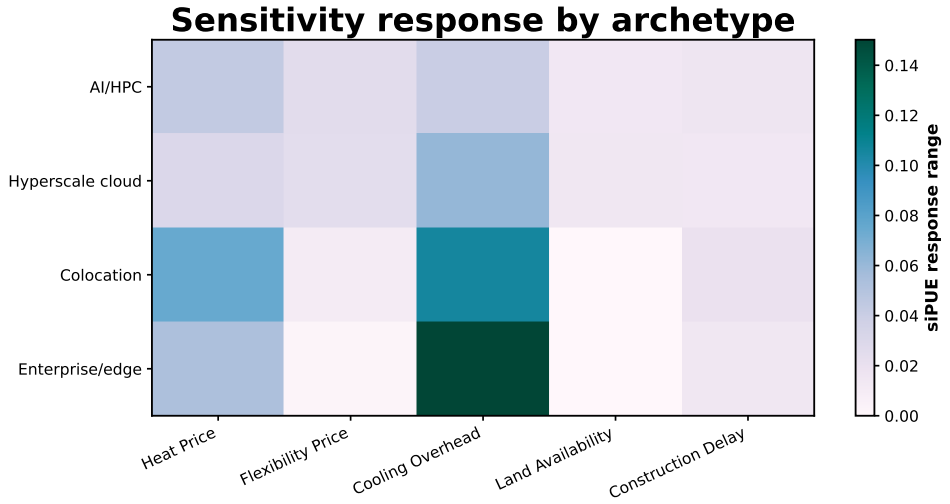


Figure 17: Sensitivity response by archetype.

4.7. Face validity against published benchmarks

This subsection checks whether modeled PUE and heat-reuse levels are plausible next to published real-project figures, with no model parameter fit to them [42, 43].

The Uptime Institute's 2024 survey reports an industry-average PUE of 1.56–1.58, leading hyperscale sites at 1.2 or lower, and new-build large colocation campuses around 1.4 [42]. The services-disabled comparator gives PUE values of 1.10 for AI/HPC, 1.15 for Hyperscale cloud, 1.26 for Colocation, and 1.36 for Enterprise/edge. These are plausible for newer integration-ready facilities, but should not be generalized to the full installed fleet.

For heat reuse, the Mäntsälä data center in Finland reuses 75% of its energy as district heat through a dedicated 4.0 MW heat pump [43]. Here, delivered useful heat divided by modeled facility electricity is 0.30–0.33 for AI/HPC and Colocation and 0.22–0.26 for Hyperscale cloud and Enterprise/edge. The published 75% figure is not confirmed to share the same numerator (delivered versus recovered heat), denominator (facility versus heat-pump electricity), or accounting boundary and period as the modeled ratio, so this is a qualitative plausibility comparison rather than a direct like-for-like benchmark. The lower modeled values are also consistent with general-purpose facilities that size heat recovery alongside flexibility, storage, land, and delay constraints, rather than a single-purpose heat-export site.

Table 15 summarizes the comparison and bounds how far the results should be extrapolated to specific sites.

Table 15: Face validity of modeled PUE and heat-reuse levels against published, real-project figures.

Quantity	Published reference	Modeled range (this study)
Hyperscale/AI PUE	1.1–1.2 at leading sites [42]	1.10–1.15
Colocation/enterprise PUE	1.4 (new-build) to 1.56–1.58 (industry average) [42]	1.26–1.36
Delivered useful heat / facility electricity	0.75 at a dedicated site (Mäntsälä) [43]	0.22–0.33 (robust design output)

4.8. Limitations and future work

The model is a controlled testbed for metric behavior, not a site-specific forecasting tool. Absolute NPV values, capacity choices, and KPI levels require project-specific data and calibration before practical use. The three-node network supports consistent DC-OPF accounting but not local interconnection assessment; representative days omit some operational extremes; and the land constraint is a comparative lease or opportunity-cost representation. Local siPUE applications should replace the independent system-cost values with local values.

These limits leave the central metric claim intact: PUE can penalize service-active designs because it observes facility electricity without observing the independent system-cost benefit those designs create. Archetype-specific evidence remains weakest for hyperscale flexibility, enterprise/edge flexibility, and enterprise/edge heat recovery.

The model excludes water use, embodied carbon, formal Renewable Energy Factor, and official ISO Energy Reuse Factor accounting [40, 29]. It also treats flexibility as reserve availability; activated scarcity events remain outside the validation scope. Future work should add observed transaction data, richer networks, water and embodied-carbon ledgers, contracting structures, reserve-activation validation, and external validation against measured heat and flexibility transactions.

5. Conclusion

Data centers are no longer just electricity loads to be minimized. Many now export waste heat, offer flexible demand, and use storage to ease pressure on local grids [11, 12], and policy increasingly expects this of them [9]. Power Usage Effectiveness cannot see any of that [13]. It treats the electricity spent on heat pumps, controllable load, and battery management as pure overhead, so a data center that invests in these services can look less efficient even as it creates real value for the surrounding energy system. This paper asked two questions that follow from that gap. First, can a companion metric reveal the local-system value of these services without abandoning PUE and without inventing credit where none exists? Second, do the design choices that make a data center robust to price and demand uncertainty pull PUE and this companion metric apart, or move them together?

The first question is answered by defining System-Integrated PUE, siPUE, a companion indicator that keeps PUE's numerator-over-IT-energy structure

but adds a single delay-adjusted, energy-equivalent credit to the facility-electricity numerator. That credit is drawn from one independent system-cost ledger that combines the heat, flexibility, peak-capacity, and carbon-damage value of a design, net of the extra electricity, network, and peak costs the same design adds. Private revenue, capital cost, and land cost stay out of it entirely, and when no services are active the credit is zero and siPUE reduces exactly to PUE. The second question is answered with a two-stage design-and-operation model that optimizes heat-exchanger, heat-pump, flexibility, and battery capacity jointly with hourly dispatch, for four data-center archetypes under ten shared uncertainty states, using representative-day operation, a three-node DC-OPF network, lifecycle delay accounting, and land-cost accounting. Heat, flexibility, peak, and carbon-damage value are never optimized in isolation; they are broken into separate lines only afterward, to show where a jointly chosen design's combined credit comes from. The model compares a services-disabled comparator, a service-active baseline, and a robust min–max-regret design against a scenario-perfect benchmark.

The results show both mechanisms at work. Weighted by annual IT energy, the services-disabled comparator, which carries the same efficiency retrofit and physical constraints as the service-active designs, gives a PUE and siPUE of 1.167. The baseline service-active design raises average PUE to 1.238 but lowers siPUE to 1.139, earning 5.603 EUR/MWh IT of delay-adjusted system benefit. The robust design pushes PUE further, to 1.343, while lowering siPUE to 1.152 and earning a larger benefit of 10.825 EUR/MWh IT. The credit is about 1.40 times the baseline facility-energy burden and about 1.08 times the robust facility-energy burden. System value keeps growing as integration intensifies, but more slowly than the electricity it costs to deliver it. At the level of individual scenarios, siPUE falls below the services-disabled comparator in 36 of 40 baseline rows and 23 of 40 robust rows, even though PUE rises in every row.

These results come with limits that matter for how they should be read. The siPUE improvement depends on how carbon damage is valued. Remove carbon damage from the ledger, and the average siPUE gain disappears; robust siPUE then rises above the services-disabled comparator rather than falling below it. The robust design was also chosen to minimize worst-case cost regret rather than worst-case siPUE regret, and that choice has a cost. Re-solving for siPUE regret directly lowers mean siPUE by much more, for a lifecycle-cost penalty under 0.11%, so the robust design reported here should be read as cost-regret robust rather than siPUE-optimal. The property, held-

out, and out-of-sample checks confirm that the metric behaves consistently and that all 12 held-out rows and all 40 synthetic out-of-sample rows remain feasible, but these are internal checks. They confirm the metric's arithmetic and the model's behavior, not its match to real installations; that still needs observed project data. Published PUE and heat-reuse figures place the modeled ranges within a plausible span for the intended archetypes [42, 43], which is a plausibility check rather than independent validation.

Taken together, these results answer both research questions, with a qualification attached to each. siPUE reveals local-system value where a jointly optimized design creates it and collapses back to PUE where none exists, which answers the first question by construction rather than by assumption. Robust design choices do widen the gap between PUE and siPUE, raising facility electricity while still lowering the system-integrated indicator, at least under the social-carbon valuation used here, which answers the second. The practical takeaway is conditional rather than universal. Some service-active data-center designs will look worse on PUE while doing real, measurable good for the local energy system, but whether that good is large enough to justify the extra electricity depends on local heat and flexibility prices and on how carbon damage is priced. PUE should keep doing what it already does well, tracking internal facility overhead [13, 40]. siPUE's role is to sit alongside it, with its reference price, delay treatment, and services-disabled comparator stated explicitly, so a reader can judge whether a facility's extra electricity is buying real local-system value.

Appendix A. Full operational model formulation

Sections 3 and 4 describe the operational model in prose; this appendix gives the complete hourly linear program each design (identity no-services, services-disabled, baseline, robust, scenario-perfect) is evaluated against, at the level needed to reproduce it. All sums over t run over the representative-day horizon $t = 1, \ldots, T$ with hourly weight w_t (Section 3); $\Delta t = 1$ h. The reference implementation, including exact parameter values, is the canonical source (Code availability, below).

A.1 Facility electricity balance and cooling

IT service power IT_t is bounded by the exogenous workload profile $\mathrm{it}_t^{\mathrm{base}}$ and, when flexibility is active, a symmetric headroom of $\pm\phi\,\mathrm{it}_t^{\mathrm{base}}$ around it, where ϕ is the archetype's flexible-load fraction. Cooling overhead is

$\mathrm{COOL}_t = \alpha_t \mathrm{IT}_t$, with α_t the (efficiency-adjusted) overhead ratio profile. Facility load is

$$E_{\mathrm{FAC},t} = \mathrm{IT}_t + \mathrm{COOL}_t + P_t^{\mathrm{HP}} + P_t^{\mathrm{serv}} + \mathrm{CH}_t - \mathrm{DIS}_t, \tag{A.1}$$

where P_t^{HP} is heat-pump electricity, $\mathrm{CH}_t, \mathrm{DIS}_t$ are battery charge/discharge power, and P_t^{serv} is parasitic service overhead (metering, controls) proportional to delivered heat, delivered reserve, and installed heat-exchanger and flexibility capacity. $E_{\mathrm{FAC}} = \sum_t E_{\mathrm{FAC},t}\,\Delta t\, w_t$ and $E_{\mathrm{IT}} = \sum_t \mathrm{IT}_t\,\Delta t\, w_t$ define $\mathrm{PUE} = E_{\mathrm{FAC}}/E_{\mathrm{IT}}$.

A.2 Heat recovery and heat-pump operation

When heat recovery is active, recovered heat H_t^{rec} splits into an IT-sourced and cooling-sourced share in proportion to α_t, is capped by both the recoverable-heat fraction of IT-plus-cooling load and by installed exchanger capacity K_{HX}, and flows to the heat pump subject to a pipe-capacity cap. Delivered heat $H_t^{\mathrm{in}} = \rho_t^{\mathrm{del}} H_t^{\mathrm{flow}}$ and heat-pump electricity $P_t^{\mathrm{HP}} = \rho_t^{\mathrm{pow}} H_t^{\mathrm{flow}} \leq K_{\mathrm{HP}}$, where $\rho_t^{\mathrm{del}}, \rho_t^{\mathrm{pow}}$ are COP- and pipe-loss-dependent source-to-delivered and source-to-power ratios. Delivered heat can be buffered in a thermal store before sale: $H_t^{\mathrm{in}} + \mathrm{DIS}_t^{\mathrm{th}} = H_t^{\mathrm{sold}} + \mathrm{CH}_t^{\mathrm{th}}$, where $\mathrm{CH}_t^{\mathrm{th}}, \mathrm{DIS}_t^{\mathrm{th}} \geq 0$ are store charge and discharge power. Stored heat follows $S_{t+1}^{\mathrm{th}} = (1-\lambda^{\mathrm{th}})S_t^{\mathrm{th}} + \eta_{\mathrm{ch}}^{\mathrm{th}}\mathrm{CH}_t^{\mathrm{th}}\Delta t - \mathrm{DIS}_t^{\mathrm{th}}\Delta t/\eta_{\mathrm{dis}}^{\mathrm{th}}$, bounded by store capacity K^{th}, with charge and discharge power each capped independently and the store cycled shut over the representative-day horizon ($S_0^{\mathrm{th}} = S_T^{\mathrm{th}} = 0$). Delivered heat, net of the store, is sold into up to K district-heating price segments, each capped by a share of district-heating acceptance capacity, giving $H_t^{\mathrm{sold}} = \sum_k Q_{k,t}$. When heat is disabled, all heat and thermal-store variables are fixed to zero.

A.3 Flexibility headroom

Load-based reserve $\mathrm{RF}_t^{\mathrm{load}}$ is bounded by installed flexibility capacity K_{FLEX} and by a fraction of the reference IT demand, and must leave enough IT headroom that $\mathrm{IT}_t + \mathrm{RF}_t^{\mathrm{load}}$ does not exceed that reference demand. Battery-based reserve $\mathrm{RF}_t^{\mathrm{bat}}$ is bounded by unused inverter capacity, $K_{\mathrm{BAT,P}} - \mathrm{DIS}_t$, and by energy above the UPS reserve floor divided by a minimum reserve-delivery duration (A.4). Total reserve is $\mathrm{RF}_t = \mathrm{RF}_t^{\mathrm{load}} + \mathrm{RF}_t^{\mathrm{bat}}$; when flexibility is disabled, both are fixed to zero.

A.4 Battery storage and UPS reserve

State of charge follows $\mathrm{SOC}_{t+1} = \mathrm{SOC}_t + \eta^{\mathrm{ch}}\mathrm{CH}_t\Delta t - \mathrm{DIS}_t\Delta t/\eta^{\mathrm{dis}}$, bounded by installed energy capacity $K_{\mathrm{BAT,E}}$, with charge/discharge power bounded by installed power capacity $K_{\mathrm{BAT,P}}$. $\mathrm{SOC}_0 = 0.5\,K_{\mathrm{BAT,E}}$ and $\mathrm{SOC}_T = \mathrm{SOC}_0$ close the cycle over the full representative-day horizon, not per individual day (Section 3). A UPS reserve floor $\mathrm{SOC}_t \geq \gamma^{\mathrm{ups}} K_{\mathrm{BAT,E}}$ holds at every hour; only energy above this floor is available for flexibility reserve.

A.5 Three-node DC-OPF network

Nodes $\{N_1, N_2, N_3\}$ (with the data center attached to one designated node) balance renewable generation $G^{\mathrm{ren}}_{n,t} \leq \bar{G}^{\mathrm{ren}}_{n,t}$, thermal generation $G^{\mathrm{th}}_{n,t}$ (capped by node-specific thermal capacity), and DC power flows $F_{12,t}, F_{23,t}, F_{13,t}$ against exogenous background load and, at the data-center node, $E_{\mathrm{FAC},t}$. Flows follow a linearized (DC) power-flow relation through nodal phase angles $\theta_{n,t}$ (with $\theta_{N_1,t} = 0$ as reference) and susceptances, and are bounded by scenario-specific line capacities. A congestion-utilization variable $U_{\mathrm{grid},t} \in [0, 1]$ tracks the maximum fractional line utilization across all three lines and enters the objective with a small coefficient so that, absent any other reason to keep it slack, the true worst-case utilization is reported rather than an arbitrary feasible upper bound.

A.6 Carbon accounting and cap

Physical emissions are $E^{\mathrm{co2}}_t =$ (thermal-generation emissions) + (baseline-boiler heat emissions on undelivered district-heating demand) − (avoided boiler emissions on delivered heat H^{sold}_t), aggregated to $E^{\mathrm{co2}} = \sum_t E^{\mathrm{co2}}_t\,\Delta t\,w_t$. A scenario-specific physical cap enforces $E^{\mathrm{co2}} \leq \mathrm{Cap}^{\mathrm{applied}}_s$ (Eq. 1) in every design case except the identity no-services and unconstrained diagnostic solves. Market carbon cost enters the private lifecycle objective at the scenario's ETS price; carbon damage enters only the independent system-cost ledger at the social cost of carbon (Table 2).

A.7 Land constraints

Land use is $\ell = \ell^{\mathrm{core}} + \ell^{\mathrm{cool}}(\alpha) + a_{\mathrm{HX}}K_{\mathrm{HX}} + a_{\mathrm{HP}}K_{\mathrm{HP}} + a_{\mathrm{FLEX}}K_{\mathrm{FLEX}} + a_{\mathrm{BAT,E}}K_{\mathrm{BAT,E}} + a_{\mathrm{BAT,P}}K_{\mathrm{BAT,P}}$, where ℓ^{core} is the existing facility footprint, $\ell^{\mathrm{cool}}(\alpha)$ scales with existing cooling capacity, and each $a_{(\cdot)}$ is a per-unit-capacity land coefficient. When a scenario land limit is set, ℓ is constrained not to exceed it; land cost enters the objective as an annualized lease or opportunity cost on ℓ.

A.8 Peak-capacity cost and construction delay

Peak import $P^{\text{peak}} \geq E_{\text{FAC},t}\ \forall t$ is the maximum facility load over the representative-day horizon (Section 3); it enters the private objective as a demand charge and the independent ledger as a peak-capacity value, both linear in P^{peak}. Construction delay does not appear as an operational constraint: it is a lifecycle scalar $a_s \in [0, 1]$ (Eq. 5) applied to the siPUE credit after the operational problem is solved, and separately to the lifecycle NPV split between pre- and post-commissioning periods.

A.9 Objective

Every design/operations solve minimizes lifecycle market cost – the present-value sum of annual resource, network, carbon-market, demand-charge, and SLA costs, net of heat and flexibility revenue and efficiency opex credit, plus capex and annualized land cost – except the diagnostic siPUE-regret and emissions-minimizing solves in Sections 4 and 3, which substitute siPUE or physical emissions as the primary objective with lifecycle cost as a lexicographic tie-break where stated.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data and results underlying this study are available in the same BRAVE GitHub repository as the code (see Code availability below).

Code availability

The computational implementation supporting this study is available in the BRAVE GitHub repository: https://github.com/sambeets/BRAVE/tree/main/sipue_modelling.

Acknowledgements

This research has received financial support from Interreg project BRAVE, and FME RenewHydro (Grant 350468, Research Council of Norway).

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